Course Outline

Bruce D. Marron

Part I
Intro to Remote Sensing in Agriculture

I. Theme: Agriculture in the 21st C.

Section: Ag's Challenges and Solutions

1. Challenges
    (a) The Anthropocene
    (b) Ecosystem Decline
    (c) Climate Change
    (d) Sustainability and Resiliency
2. Solutions
    (a) Agroecology and Regenerative Ag
    (b) Urban Ag
    (c) Tribal Knowledge
    (d) Advanced Crop Breeding
    (e) Adaptive Management
    (f) Tech

Section: Remote Sensing Tools

1. Satellites
    (a) Sential
    (b) Satellite Data
    (c) Satellite Maps
2. Drones
    (a) Drones
    (b) Drone Data
    (c) Drone Maps

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II. Theme: Cloud Computing in the 21st C.

Section: Introduction to Professional AI and API Use

1. Cloud, API, and AI Service Providers
    (a) Anthropic
    (b) Google
        i. Google Cloud
        ii. Google Eart Engine 2 Copernicus
2. Accessing AIs and APIs
    (a) Python
    (b) Spyder
    (c) SDKs

III. Theme: Records Management in the 21st C.

Introduction to Records and Document Management

1. QAQC for Docs
2. GitHub

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